

Welcome to Clever Cluster

In this app, you can use pre-loaded data or your own to explore unsupervised machine learning through Principal Components Analysis, K-Means Clustering, and Hierarchical Clustering!

K-Means Model Evaluation

There are a few ways we can evaluate our K-Means model:

- Silhouette Score: This score indicates how similar an item is to its own cluster compared to other clusters (higher numbers are better)
- Accuracy: This is a measure of the proportion of items that are clustered properly given the user's target variable
- Within-Cluster Sum of Squares: This indicates how similar items in a cluster are to each other

Silhouette Score

0.43

Accuracy

0.13

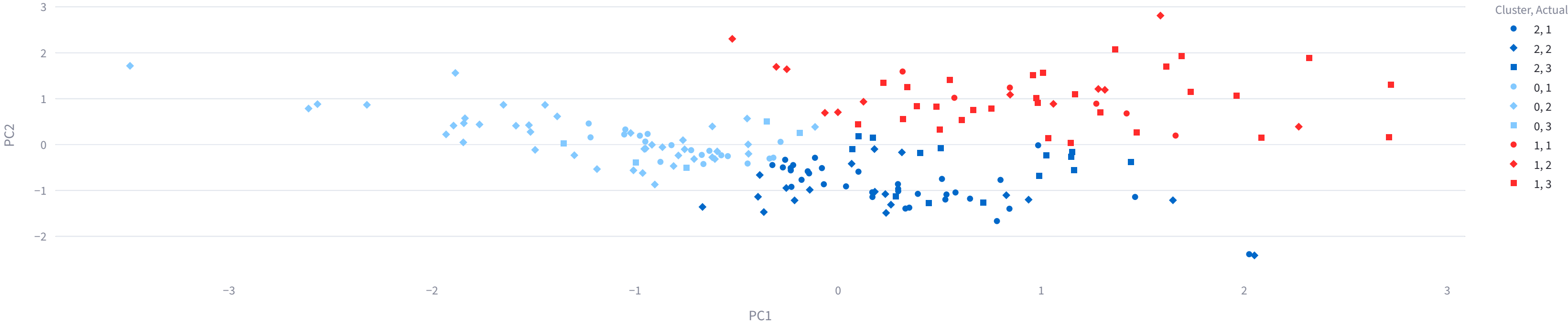
Within-Cluster Sum of Squares

126.75

Using PCA to visualize clusters

Using PCA, we can condense feature variables into two components, which, while not perfect, allows us to visualize the K-Means clusters, as well as how they compare to actual label values.

PCA: K-Means Clusters vs Real Labels

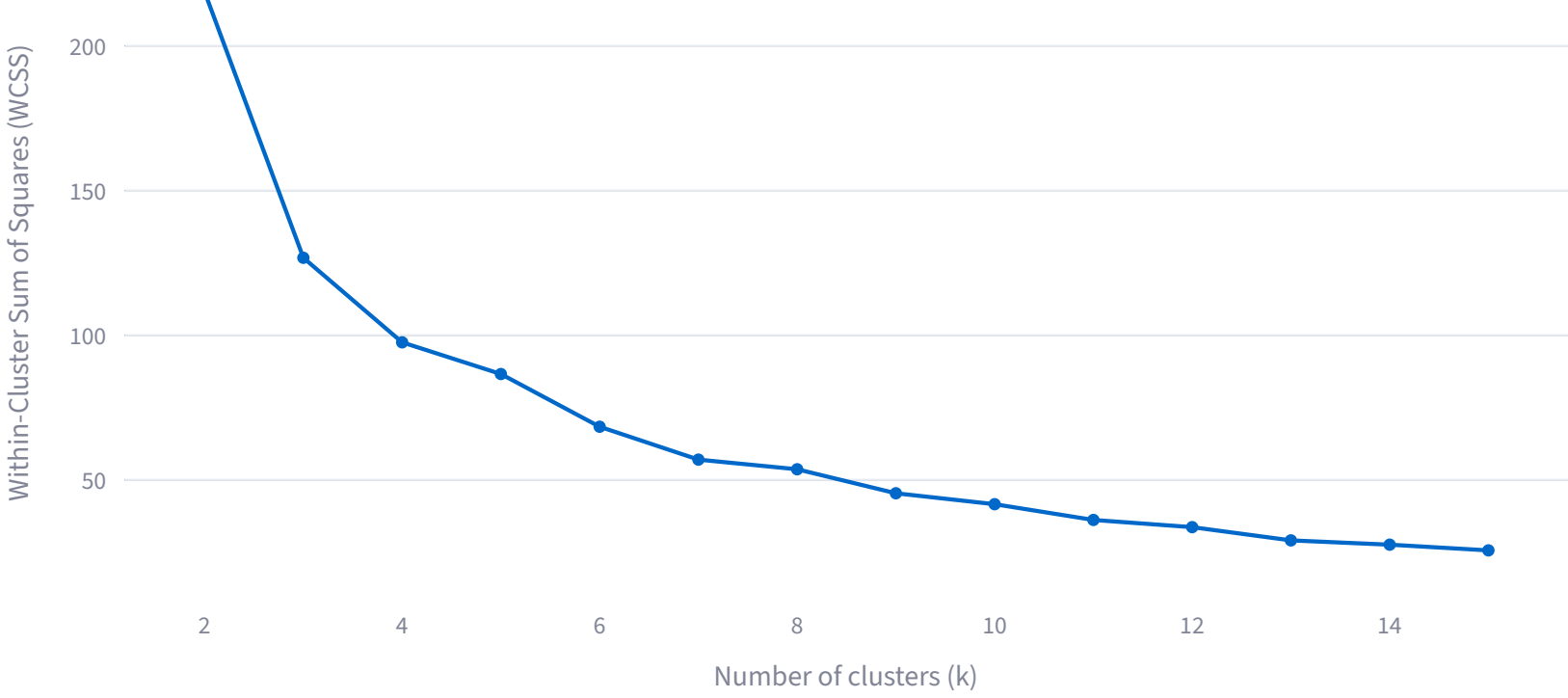


Choosing Optimal Number of Clusters

There are two easy visual ways to choose the optimal number of clusters (k)!

1. **Elbow Plot** - This plots WCSS vs. number of clusters, and the optimal k is at the 'elbow', or where the rate of decrease sharply changes
2. **Silhouette Plot** - This plots silhouette score vs. number of clusters, and the optimal k is the highest point

Elbow Method for Optimal K



Silhouette Score for Optimal K

